

1. ADD Of two numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code window contains the following instructions:

```
1 LDA 8500
2 MOV B,A
3 LDA 8501
4 ADD B
5 STA 8502
6 RST 1
```

The registers window shows the following values:

Register	Value
A	46
BC	1E 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0C
SP	FF FF
Int-Reg	00

The flag window shows the following values:

Flag	Value
S	0
Z	0
AC	1
P	0
C	0

The memory window shows the following values:

Address (Hex)	Address	Data
2134	8500	30
2135	8501	40
2136	8502	70
2137	8503	0
2138	8504	0
2139	8505	0
213A	8506	0
213B	8507	0
213C	8508	0
213D	8509	0
213E	8510	0
213F	8511	0

The assembler message window shows the following message:

```
Line No | Assembler Message
0 | Program assembled successfully
```

2. SUB of two numbers

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code window contains the following instructions:

```
1 LDA 7500
2 MOV B,A
3 LDA 7501
4 SUB B
5 STA 7502
6 RST 1
```

The registers window shows the following values:

Register	Value
A	14
BC	0A 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0C
SP	FF FF
Int-Reg	00

The flag window shows the following values:

Flag	Value
S	0
Z	0
AC	0
P	1
C	0

The memory window shows the following values:

Address (Hex)	Address	Data
1D4C	7500	10
1D4D	7501	30
1D4E	7502	20
1D4F	7503	0
1D50	7504	0
1D51	7505	0
1D52	7506	0
1D53	7507	0
1D54	7508	0
1D55	7509	0
1D56	7510	0
1D57	7511	0

The assembler message window shows the following message:

```
Line No | Assembler Message
0 | Program assembled successfully
```

3. Addition of 2 numbers with carry

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code window contains the following instructions:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 STC
5 ADC B
6 STA 2502
7 HLT
```

The registers window shows the following values:

Register	Value
A	14
BC	0A 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 0C
SP	FF FF
Int-Reg	00

The flag window shows the following values:

Flag	Value
S	0
Z	0
AC	0
P	1
C	0

The memory window shows the following values:

Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	15
09C6	2502	41
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message window shows the following message:

```
Line No | Assembler Message
0 | Program assembled successfully
```

4. Sub with borrow

The screenshot shows an 8085 assembly simulator. The assembly code is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 MOV C,A
5 MOV A,B
6 STC
7 SBB C
8 STA 2502
9 HLT
```

The memory window on the right shows the following data:

Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	15
09C6	2502	9
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message window shows: "Program assembled successfully".

5. Multiplication of 2 numbers

The screenshot shows an 8085 assembly simulator. The assembly code is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 MOV C,A
5 MVI A,00H
6
7 LOOP: ADD B
8 DCR C
9 JNZ LOOP
10
11 STA 2502
12 HLT
```

The memory window on the right shows the following data:

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	4
09C6	2502	20
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message window shows: "Program assembled successfully".

6. Division of two numbers

The screenshot shows an 8085 assembly simulator. The assembly code is as follows:

```
1 LDA 2500
2 MOV B,A
3 LDA 2501
4 MOV C,A
5 MOV D,00H
6
7 LOOP: MOV A,B
8 CMP C
9 JNC DONE
10 SUB C
11 MOV B,A
12 INR D
13 JMP LOOP
14
15 DONE: MOV A,D
16 STA 2502
17
18 MOV A,B
19 STA 2503
20
21 HLT
```

The memory window on the right shows the following data:

Address (Hex)	Address	Data
09C4	2500	11
09C5	2501	2
09C6	2502	5
09C7	2503	1
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message window shows: "Program assembled successfully".

7. Greatest of two numbers

The screenshot shows an 8085 assembly program in a simulator. The program is designed to find the greatest of two numbers stored at memory locations 2500 and 2501. It uses the JC (Jump if Carry) instruction to branch to a label 'FIRST' if the number at 2500 is greater than the number at 2501. Otherwise, it proceeds to a label 'SECOND'.

Assembly Code:

```
1 LDA 2500
2 MOV B,A
3
4 LDA 2501
5 CMP B
6
7 JC FIRST
8
9 STA 2502
10 HLT
11
12 FIRST: MOV A,B
13 STA 2502
14 HLT
```

Memory Dump:

Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	35
09C6	2502	35
09C7	2503	1
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Assembler Message:

```
Line No Assembler Message
0 Program assembled successfully
```

8. Smallest of two numbers

The screenshot shows an 8085 assembly program in a simulator. The program is designed to find the smallest of two numbers stored at memory locations 2500 and 2501. It uses the JC (Jump if Carry) instruction to branch to a label 'SECOND' if the number at 2500 is greater than the number at 2501. Otherwise, it proceeds to a label 'FIRST'.

Assembly Code:

```
1 LDA 2500
2 MOV B,A
3
4 LDA 2501
5 CMP B
6
7 JC SECOND
8
9 MOV A,B
10 STA 2502
11 HLT
12
13 SECOND: STA 2502
14 HLT
```

Memory Dump:

Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	15
09C6	2502	15
09C7	2503	1
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Assembler Message:

```
Line No Assembler Message
0 Program assembled successfully
```

9. Swapping of two numbers

The screenshot shows an 8085 assembly program in a simulator. The program is designed to swap the numbers stored at memory locations 2500 and 2501. It uses the MOV instruction to move the value from 2500 to register C, then from 2501 to register A, and finally from register C back to memory location 2500.

Assembly Code:

```
1 LDA 2500
2 MOV B,A
3
4 LDA 2501
5 MOV C,A
6
7 MOV A,B
8 STA 2501
9
10 MOV A,C
11 STA 2500
12 HLT
```

Memory Dump:

Address (Hex)	Address	Data
09C4	2500	15
09C5	2501	25
09C6	2502	0
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Assembler Message:

```
Line No Assembler Message
0 Program assembled successfully
```

10. 1's Complement

The screenshot shows an 8085 assembler interface. The assembly code is as follows:

```

1 LDA 2500
2 CMA
3 STA 2501
4 HLT
  
```

The memory dump on the right shows the program being assembled at address 2500:

Address (Hex)	Address	Data
09C4	2500	55
09C5	2501	200
09C6	2502	0
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message at the bottom states: "Program assembled successfully".

11. 2's complement

The screenshot shows an 8085 assembler interface. The assembly code is as follows:

```

1 LDA 2500
2 CMA
3 INR A
4 STA 2501
5 HLT
  
```

The memory dump on the right shows the program being assembled at address 2500:

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	251
09C6	2502	0
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message at the bottom states: "Program assembled successfully".

12. Even or Odd of the given number

The screenshot shows an 8085 assembler interface. The assembly code is as follows:

```

1 LDA 2500
2 ANI 01H
3 JZ EVEN
4 MVI A, 01H
5 STA 2501
6 HLT
7 EVEN: MVI A, 00H
8 STA 2501
9 HLT
  
```

The memory dump on the right shows the program being assembled at address 2500:

Address (Hex)	Address	Data
09C4	2500	9
09C5	2501	1
09C6	2502	0
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message at the bottom states: "Program assembled successfully".